

OLIVIA A. M. MARTIN

Curriculum Vitae

Email — Website — LinkedIn — Google Scholar

RESEARCH INTERESTS

- Nuclear quantum effects, water-mineral interfaces, hydrogen bonding structure and dynamics, computational spectroscopy
- Membrane-mineral interfaces, thermodynamic theory, prebiotic chemistry, origins of life
- Molecular dynamics simulations, path integral molecular dynamics, enhanced sampling techniques, machine learning

EDUCATION

Ph.D. in Chemistry, Physical Chemistry

Advisor: Dr. Richard C. Remsing

Expected May 2027

Rutgers University, Piscataway, NJ

B.S. in Chemistry, *cum laude*

Advisor: Dr. Nathan C. Eddingsaas

May 2022

Rochester Institute of Technology, Henrietta, NY

Minor: Environmental Sciences

RESEARCH EXPERIENCE

Graduate Student Researcher

June 2022 - Present

Advisor: Dr. Richard C. Remsing

Department of Chemistry and Chemical Biology, Rutgers University, Piscataway, NJ

Emerson Summer Undergraduate Research Fellowship

May - August 2021

Advisor: Dr. Nathan C. Eddingsaas

College of Science, Rochester Institute of Technology, Henrietta, NY

Undergraduate Student Researcher

August 2018 - May 2022

Advisor: Dr. Nathan C. Eddingsaas

Department of Chemistry and Materials Science, Rochester Institute of Technology, Henrietta, NY

HONORS AND AWARDS

2022 Undergraduate Senior Chemistry Achievement Award, *American Institute of Chemists*

2021 Nathaniel Rochester Society Scholarship

2021 Good Citizenship Award, *Rochester Institute of Technology*

2020 Nathaniel Rochester Society Scholarship

2020 Good Citizenship Award, *Rochester Institute of Technology*

2020 Alumni Endowed Academic Scholarship, *College of Science, Rochester Institute of Technology*

2020 Outstanding Second Year Student Award, *Rochester Institute of Technology*

PUBLICATIONS

1. "A bellwether for microplastic in wetland catchments in the Great Lakes region" Dominique S. Gilbert, Brett A. Hayhurst, Sarah Grubisich, Nick Schneider, **Olivia Martin**, Christopher DeNyse, Kristina M. Chomiak, Anna Christina Tyler, Nathan C. Eddingsaas. *Journal of Great Lakes Research*, **2024**, 50, 102411
DOI: 10.1016/j.jglr.2024.102411

SELECTED TALKS AND PRESENTATIONS

Oral Presentations

7. "Molecular and Thermodynamic Insights into Water and Amphiphiles at the α -Alumina Interface", Water and Aqueous Solutions Gordon Research Seminar 2026, Holderness, NH
6. "Molecular and Thermodynamic Insights into Prebiotic Lipid Assembly", Astrobiology Science Conference 2026, Madison, WI

5. "Lipid Assembly Formation on Minerals: A Path to Protocells", Astrobiology Graduate Conference 2025, Boulder, CO
4. "Thermodynamics of Primitive Lipid Assemblies at Mineral Surfaces", Astrobiology Graduate Conference 2024, Ithaca, NY
3. "Interfacial Structure and Vibrational Relaxation of Aqueous Solutions near Alumina", American Chemical Society Mid-Atlantic Regional Meeting (MARM) 2024, State College, PA
2. "Analytical Methods and Processes of Environmental Microplastic", RIT Undergraduate Research Symposium 2021, Rochester, NY
1. "The chemical and physical changes to plastics aged in surface and benthic freshwater aquatic systems", State of Lake Ontario Conference 2021, Rochester, NY

Poster Presentations

8. "Molecular and Thermodynamic Insights into Water and Amphiphiles at the α -Alumina Interface", Water and Aqueous Solutions Gordon Research Conference 2026, Holderness, NH
7. "Structure and Dynamics of Water and Amphiphiles at the α -Alumina Surface", Chemistry and Physics of Liquids Gordon Research Conference 2025, Holderness, NH
6. "Structure and Dynamics of Water and Amphiphiles at the α -Alumina Surface", Chemistry and Physics of Liquids Gordon Research Seminar 2025, Holderness, NH
5. "Thermodynamics of Primitive Lipid Assemblies at Mineral Surfaces", Chemistry in Solution and at Interfaces Workshop: Deep Modeling for Molecular Simulations Workshop 2024, Princeton, NJ
4. "Fatty Acid Assembly at Water-Mineral Interfaces, Water and Aqueous Solutions", Water and Aqueous Solutions Gordon Research Conference 2024, Holderness, NH
3. "Fatty Acid Assembly at Water-Mineral Interfaces, Water and Aqueous Solutions", Water and Aqueous Solutions Gordon Research Seminar 2024, Holderness, NH
2. "Thermodynamics of Primitive Lipid Assemblies at Mineral Surfaces", Astrobiology Science Conference 2024, Providence, RI
1. "Microplastic Pollution in the Benthos of Rural and Urban Tributaries of Southern Lake Ontario", Joint Aquatic Sciences Meeting 2022, Grand Rapids, MI

PROFESSIONAL SERVICE

Co-chair	2025 - Present
Chemistry and Physics of Liquids Gordon Research Seminar 2027	Holderness, NH
Co-chair	2025 - Present
Astrobiology Graduate Conference 2026	Tucson, AZ
Discussion Leader	2025
Chemistry and Physics of Liquids Gordon Research Seminar 2025	Holderness, NH
Director of Outreach	2023 - 2025
Chemistry and Chemical Biology Graduate Student Organization	Rutgers University, Piscataway, NJ
Executive Board	2023 - 2025
Treasurer, Supporting Parents and Caregivers at Rutgers Organization	Rutgers University, Piscataway, NJ
Volunteer	2019 - 2022
Women in Science	Rochester Institute of Technology, Henrietta, NY
Representative	2020 - 2021
Young Chemists Committee, American Chemical Society	Rochester Institute of Technology, Henrietta, NY
President	2019 - 2021
American Chemical Society	Rochester Institute of Technology, Henrietta, NY

Student Ambassador College of Science Student Advisory Board	2018 - 2021 Rochester Institute of Technology, Henrietta, NY
Volunteer Rochester Museum and Science Center	2020 Rochester, NY
Event Coordinator American Chemical Society	2019 Rochester Institute of Technology, Henrietta, NY
Chair Hospitality, 64th Annual Undergraduate Research Symposium, American Chemical Society	2019 Rochester, NY

TEACHING AND ADVISING

Graduate Teaching Assistant Organic Chemistry I Lab	2024 - Present Rutgers University, Piscataway, NJ
Graduate Mentor Remsing Lab	2024 - 2025 Rutgers University, Piscataway, NJ
Tutor Organic Chemistry I and II	2022 - 2023 Rutgers University, Piscataway, NJ
Undergraduate Mentor College of Science	2021 - 2022 Rochester Institute of Technology, Henrietta, NY
Undergraduate Teaching Assistant General and Analytical Chemistry Lab	2020 - 2022 Rochester Institute of Technology, Henrietta, NY
Learning Assistant General Chemistry I and II	2020 - 2021 Rochester Institute of Technology, Henrietta, NY